

SEMI-AUTOMATIC SOLDER PRINTER

Touch Panel Operation Instruction Manual



SEMI-AUTOMATIC HIGH PRECISION SOLDER PRINTER

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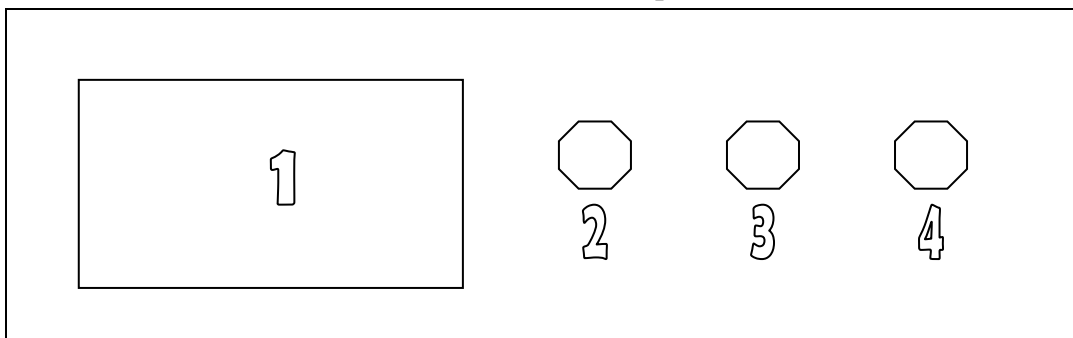
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1. Futures Introduction:

- 1) Printing plates and steel from the level of interconnection with precision fine-tuning rod adjustment.
- 2) Machine arm can be adjusted left and right, respectively, for 370 * 470mm, 420 * 520mm, 550 * 650mm different steel mesh sizes.
- 3) Combined printing platen grooves and positioning PIN, set simple, quick facelift, and is suitable for single-sided and double-sided board substrate production.
- 4) The way the school board adopted the arm (steel mesh) to move with convenient print (platen) X, Y, Z calibration adjustments.
- 5) Electrical action using PLC, inverter control, touch screen interface, the freedom to choose a single / double times squeegee and manual and automatic operation mode, such as the use of day-precision motor and linear guide rail combination, make printing more stable scraper seat .
- 6) Double blade of printing pressure respectively using precision throttle back and down to adjust the blade lift cylinder can be fast or slow, and can effectively avoid resonance.
- 7) Printing seat lift can lift 45 ° upwards and fixed, while taking advantage of the cleaning blade loading and steel nets.
- 8) Printing seat can be moved forward to the position with a fixed pattern of steel version, in order to get better the printed results.
- 9) Printing seat twin blade of the level can be set freely by adjusting the nut.

2. Electric Panel Description



- 1) Touch screen operator interface.
- 2) Printing potentiometer.
- 3) Emergency button.
- 4) Power switch

3. Machine Operation and Adjustment

1) Set the essentials:

By steel clamp arm up down keys, the steel clamp arm down to the starting point under the PCB placed on the platen presses at the top of the wheel rotational movement, according to the PCB thickness adjustment pitch stencil printing table with plates.

2) Support PIN setting mode:

Positioning the machine's PIN movable Please follow the outline graphic stencil PIN will be placed on the platen support moderate position.

3) Network setting mode:

After the completion of the action, and then printing stencil printing PCB proofing methods are as follows: The PCB placed on the support PIN, PIN positioning with positioning firm (PCB positioning at least three corners), put the stencil on the pads with the PCB, after mobile steel clamp arm will stencil folder in prison, then the XY PCB platen adjustment screw for final fine-tuning.

4) Brush Stroke setting mode:

After the completion of PCB set of bits, and then use the second sensor switch only after printing blocks, respectively, according to the pattern of the stencil printing size is set around sensor switch position.

5) Knife assembly methods:

Scraper mining center screw type assembly, in accordance with the scraper blade insert screws to seat center.

6) Knife height adjustment:

Use of the upper blade adjustment screw to adjust the blade up and down spin out height.

7) Knife balance adjustment mode:

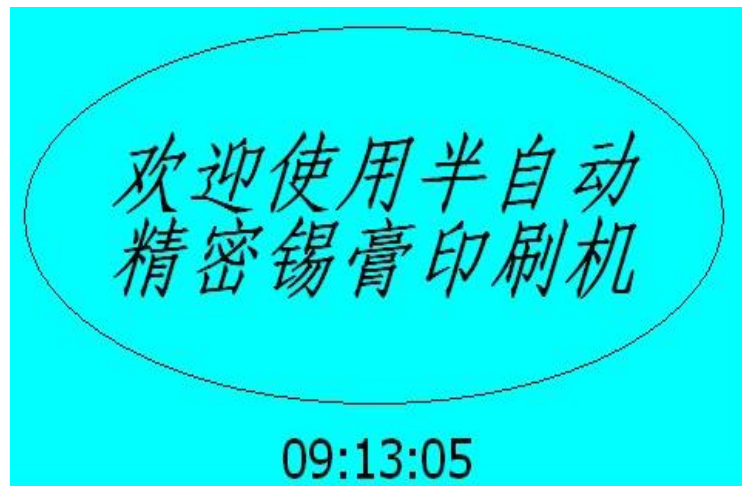
Front and rear side have a screw scraper balance control, front and rear screws need to complement each other before and after use to control the level of the blade.

8) The degree of the setting:

Under the premise of the program can be completed after production requirements, the printing speed will be lowered as far as possible to make up to the best printing results, including printing speed and travel speed steel blade stripping speeds.

4. Touch Panel Operation

1) Press the power control switch, the power display screen:



Click to enter touch-screen operator interface



- ① Press the "test mode" to enter the test bar;
- ② Press the "Jog Mode" button to enter the manual work screen;
- ③ Press the "semi-automatic mode" to enter the semi-working screen;
- ④ Press the "automatic mode" to enter the automatic working screen;
- ⑤ Press the "Settings" button to set the working screen to enter data;
- ⑥ Note: In determining the test parameters must be set before work, or can not work properly, (factory setting)

2) Testing screen

In the mode selection screen, press "Test Mode" the following screen:



In test mode by pressing the "Start" button is pressed double "start" button, Enter the test mode, the blade enters continuously.

3) Jog work interface

In the mode selection screen, press "Jog Mode" the following screen:



- ① Press the left key blade knife next move, then press the button to move the knife blade on the left;
- ② Press the right key blade knife next move, then press the button to move the right knife blade;
- ③ Printing Printing press the left button to the left jog;
- ④ Printing Printing press the right button to the right jog;
- ⑤ Press the Start button on the printing stencil, stencil pressed again.
- ⑥ Press the Back button to return to the previous screen.

4) Semi-automatic work interface

In the mode selection screen press the "semi-automatic mode" to display the following screen:



① Count: The number of production will be displayed in the upper left of the screen;

② Press the double "Start" button to start the printing presses a stroke, the press stops without pressing the pair behind the "Start" button will stop working, and then press the double "start" button presses and then start the trip; emergency case pressing the "emergency stop" button will automatically alert the next. The doctor is not (about) original position, press the "reset" button scraper back to the left of the original position

③ Press the "Return" key to return to the previous screen.

5) Automatic work interface:

In the mode selection screen, press "automatic mode" to display the following screen:



① Count: The number of production will be displayed in the upper left of the screen;

② Press the double "Start" button to start printing presses,

③ Scraper as "round-trip" a trip. Case of emergency, press the "emergency stop" button will automatically alarm. Such as when the blade

is not (about) original position, press the "Reset" button scraper back to the left of the original position.

- ④ Press the "Return" key to return to the previous screen.

6) Set the working screen

Press the Setup button in the interface to display the following screen;



- ① "All action to stop" automatic stencil work residence time, usually after a set up is set to press the Enter key;

- ② "Left parked" right time to the right to stay scraper fall line, set as above, the unit is 1S (s);

- ③ "Right stop," the left and right blade retention time down the line, is set as above; unit is 1S (s);

- ④ Press the "Return" key to return to the previous screen.

5. Main Parts Sheet

Item No.	Part No.	Description
1	PLC 编程器(CPU) PLC Programmer	Taiwan VIGOR
2	人机界面 Operation Interface	China XIANKONG
3	开关电源 Switching Power Supply	China MINGWEI
4	刮刀气缸 Scraper Cylinder	SMC
5	电磁阀 Solenoid Valve	China RENHE
6	直线导轨	Taiwan TBI

	Linear Guide	
7	感应器 Inductor	China PUBANG
8	调速阀 Speed Control Valve	JSCC
9	电机 Motor & Driver	VTV
10	继电器 Relay	Japan OMRON
11	三点组合过滤器 Three-point Combination Filter	China RENHE
12	变频器 Converter	JSCC

6. Troubleshooting

Machines such as occurred in the operation to repair the fault, please follow these troubleshooting methods

Fault Display	Fault Condition	Cause and Remedy
1	1, unable to rise 2, unable to drop	1, input or insufficient pressure (normal pressure should be maintained at 4kg-6kg); 2, the sensor is not sensing or loss, short-term, down solenoid valve failure; 3, the function has not been selected; 4, bad cylinder control valve or improper adjustment.
2	2, unable to L-R printing	2, fixed sensor is not sensing or damaged, broken.
4	1, right after the fall scraping gum can not print; 2, left printing; 3, the left screen printing can not stop will not be able to rise to the left when the sentinel	1, Left sensor is not sensing or loss, break; 2, Left and right drive motor power switch is not open or broken drive failure, left-printing speed adjuster to adjust bad or damaged; 3, Left line drive relay is damaged or broken;

5	Screen down without printing to rise	1, around the sensor is damaged or shorted; 2, the sensor is inverter interference, then the negative supply and OUPUT sensor of the capacitor in parallel with a 0.1μF.
6	No power input	1, the power supply is disconnected or bad; 2, the switch is wearing flower or bad; 3, the fuse has burned;
7	Inverter unstable to start or drive motor	1, An inverter parameter setting error; 2, the drive is damaged;
8	Squeegee unable to rise or fall	1, the cylinder drive solenoid valve failure; 2, cylinder control valve is bad or improper adjustment;

7. I/O Sheet

INPUT	INSTRUCTION	INPUT	INSTRUCTION
X0	Large Cylinder Upper Limit Induction Switch	Y0	Large Cylinder Rise-Fall
X1	Large Cylinder Lower Limit Induction Switch	Y1	Motor Coronation
X2	Left Limit Induction Switch	Y2	Motor Inversion
X3	Right Limit Induction Switch	Y3	Left Small Cylinder Rise-Fall
X4	Emergency Stop Switch	Y4	Right Small Cylinder Rise-Fall
X5	Start Button	Y5	Buzzer
X6		Y6	

8. Circuit Diagram

